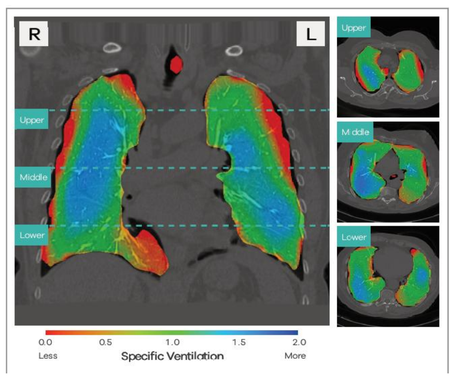
LungLogic: Intelligent Airflow Control for Realistic Breathing Simulation

ME-12-TYIN-007 | Yuqiao Li, Chun Yin Ku (John), Jiaquan Zhong | Supervisors: Jonathan Eden, Ying Tan | Mechatronics Engineering

1. Introduction

1.1 Project need

Dynamic lung phantoms provide repeatable breathing inputs for imaging and ventilation research.

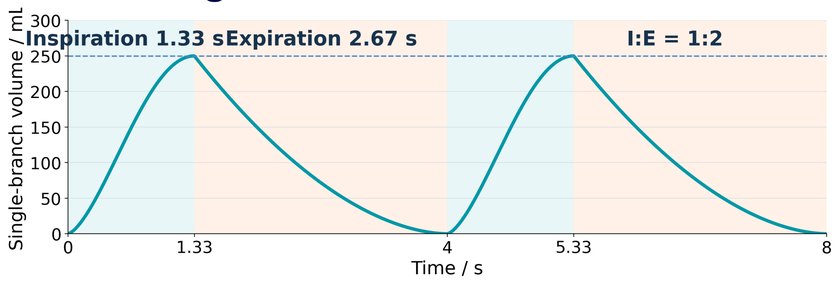


1.2 Project objective

LungLogic controls airflow in a biomimetic lung model. This stage validates one 250 mL branch before bilateral integration.

Regional ventilation imaging. Application context.

1.3 Nominal breathing reference



Breathing-cycle illustration at 15 bpm and I:E = 1:2.

RR · respiratory rate	12–16 bpm; nominal 15 bpm
I:E · inhale:exhale time	1:2; 1.33 s / 2.67 s at 15 bpm
VT · tidal volume	250 mL per branch
Repeatability	VT error within ±5%

Nominal design values, not measured performance. Target: 250 mL per branch and VT error within ±5%.

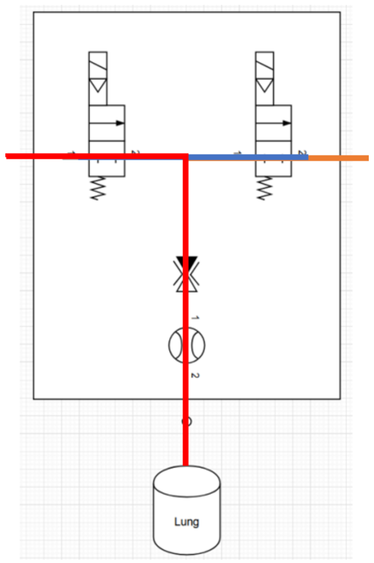
Sources: 4DMedical project scope (19 March 2026), pp. 5, 13. Ventilation image: supplied reference.

2. Pneumatic System & Hardware

2.1 Objective

Control one lung branch and measure airflow for closed-loop regulation and calibration.

2.2 Pneumatic system prototype



Prototype photograph (to be added)

Fig. 1. Single-lung route; prototype photograph pending.

2.3 Embedded control architecture

PC / Python UI → **Controller** → **24 V bridge PCB** → **Valves + sensor**

The controller runs the 100 Hz breathing and flow loop. The bridge connects logic to the routing valves, PPV and SFM3300 sensor.



Fig. 2. Control architecture and bridge PCB.

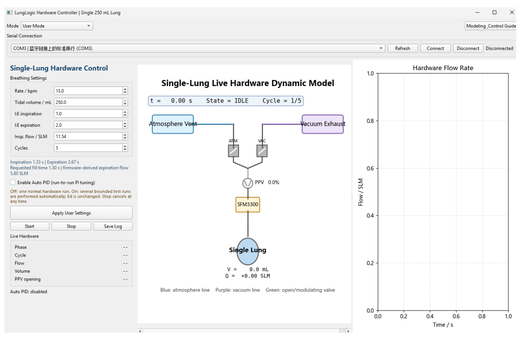
3. Integrated Control System & Software

3.1 Objective

Track flow and tidal volume with a visible, repeatable closed-loop system. Feedforward supplies the nominal valve command; sensor feedback corrects error.

3.2 Operator and developer interface

User Mode



Developer Mode

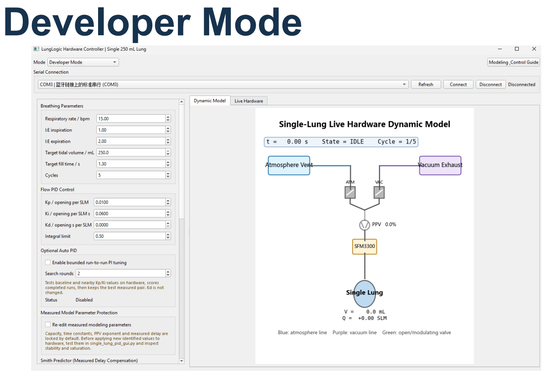


Fig. 3. User and Developer modes.

3.3 Model-assisted flow feedback

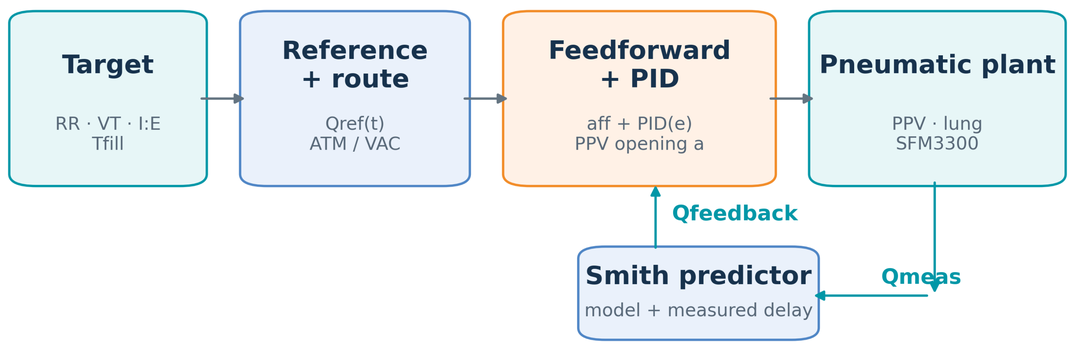


Fig. 4. Feedback control with optional Smith prediction.

3.4 Optional Auto PID and measured-delay compensation

Auto PID compares bounded Kp/Ki candidates over complete runs using flow error, tidal-volume error and valve saturation. An optional Smith predictor uses the identified plant model and measured feedback delay to estimate current flow.

4. Results & Future Work

4.1 Experimental results

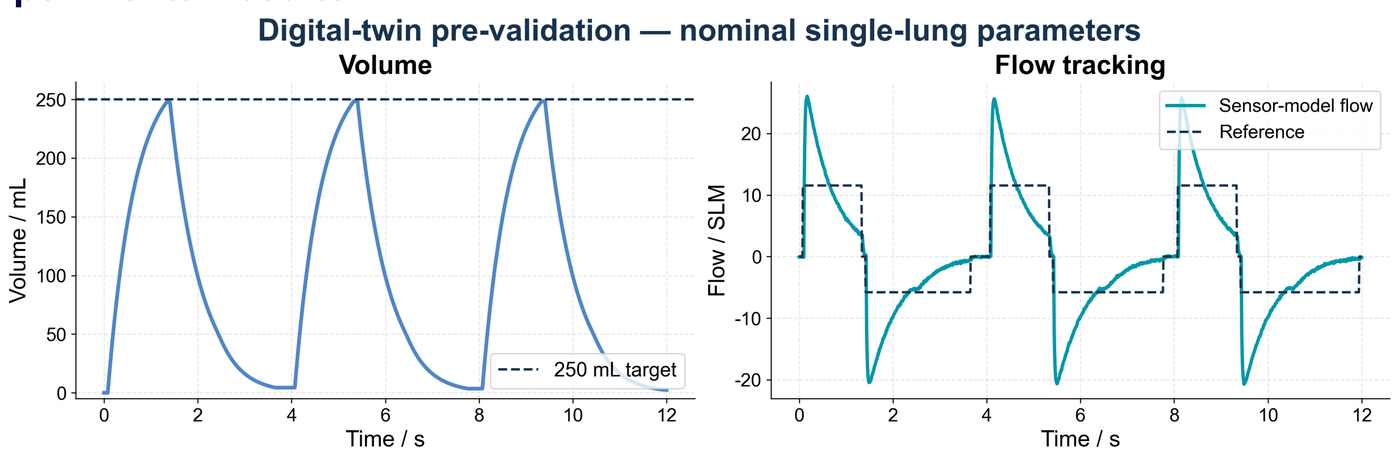


Fig. 5. Digital-twin simulation at 15 bpm (not measured hardware).

4.2 Future work

1. **Integrate the 250 mL branch**
2. **Re-identify model and delay**
3. **Validate branch repeatability**

Connect the single branch to the biomimetic lung model and test the required breathing range.